ANISH TIWARI

AI/ML Engineer | Computer Vision | Deep Learning

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PROFESSIONAL SUMMARY

B.Tech Computer Science student (2027) targeting AI/ML and computer vision roles. Developing an AI-based frame interpolation system for satellite/time-series imagery and proposing a **YOLOv8 + ResNet50** pipeline for Indian cattle and buffalo breed recognition. Python and C++ programmer with **250+ DSA problems solved on LeetCode** and Generative AI fundamentals (LLMs, RAG, LangChain).

EDUCATION

B.Tech, Computer Science & Engineering 2023 – 2027
Sri Vaishnavi College of Engineering CGPA: 7.9 / 10

Senior Secondary (Class XII), MPBSE – Science 2023
Sri Ayyappa Public Higher Secondary School Percentage: 85.6%

Secondary (Class X), MPBSE 2021
Sri Ayyappa Public Higher Secondary School Percentage: 72.8%

TECHNICAL SKILLS

Programming Languages: Python, C++, C, JavaScript, TypeScript

AI/ML & Computer Vision: Machine Learning, Deep Learning, Computer Vision, Transfer Learning, YOLOv8, ResNet50, Supervised & Unsupervised Learning, Linear Regression, Data Cleaning, Pandas, NumPy

Generative AI: LLMs, Prompt Engineering, RAG (Retrieval-Augmented Generation), Embeddings, Vector Databases, AI Agents, LangChain, LangGraph

DSA & Core CS: Data Structures & Algorithms (250+ problems solved), OOP, DBMS, Computer Networks, Operating Systems, System Design Fundamentals

Databases & Tools: SQL, MySQL, MongoDB, Git, GitHub, Linux

PROJECTS

Image-Based Breed Recognition of Indian Cattle and Buffaloes **Proposed**
Tech: *Python, Computer Vision, YOLOv8, ResNet50, Transfer Learning, Streamlit*

- **Problem:** Visually identifying Indian cattle and buffalo breeds consistently is difficult because many breeds differ only in fine-grained features, which makes breed recognition a hard image classification task.
- **Proposed approach:** a two-stage pipeline in which **YOLOv8** detects the animal and extracts the region of interest, and a **ResNet50** transfer-learning model classifies the breed.
- **Planned evaluation:** precision, recall, F1-score, and confusion matrices to study per-breed performance and confusion between similar breeds.
- **Planned prototype and use cases:** a Streamlit web interface for uploading an image and viewing the predicted breed, intended to support livestock management and agricultural applications.
- **Status:** proposed in a project synopsis; implementation, training, and results are pending.

AI-Based Frame Interpolation, Video Generation & Display System **Ongoing**
Tech: *Python, AI/ML, Computer Vision, Deep Learning*

- Developing an AI/ML-based system that generates intermediate frames between time-separated images, applied to satellite/time-series imagery.
- Applying computer vision and deep learning techniques to improve temporal continuity and produce smoother visual transitions between frames.
- **Role:** AI/ML development, as part of an ongoing academic project.

ACHIEVEMENTS & LEADERSHIP

- Solved **250+ Data Structures & Algorithms problems** on LeetCode (profile).
- Served as **Class Representative for 2 years**, developing leadership, coordination, and communication skills across a large student group.
- Actively participated in running events, building discipline, focus, and a competitive mindset alongside academic work.